

1. For the following data construct a Stem-Leaf diagram, analyze for all possible aspects, and sketch the corresponding the Box-Whisker plot.

15, 9, 15, 17, 21, 32, 16, 11, 14, 19

Raw plot

Stem	Leaf
1	5 5 7 6 1 4 9
0	9
2	1
3	2

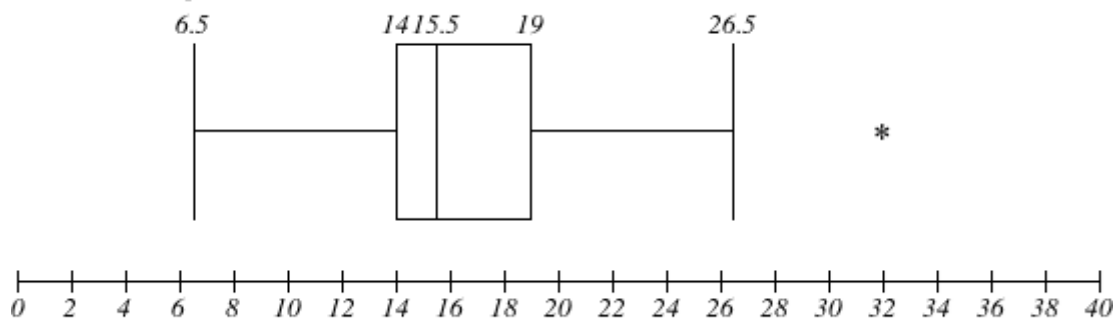
Ordered plot

Stem	Leaf
0	9
1	1 4 5 5 6 7 9
2	1
3	2

Key: 3|2 → 32

Sample size	10
Range	$32 - 9 = 23$
Semi-range	$\frac{23}{2} = 11.5$
Modal stem	1
Mode	15
Mean	$\frac{9 + 11 + 14 + 15 + 15 + 16 + 17 + 19 + 21 + 32}{10} = 15.3$
Median	$\frac{15 + 16}{2} = 15.5$
Q1	14
Q3	19
IQR	$19 - 14 = 5$
Q1~Median	1.5
Median~Q3	3.5
Nature	Right-skewed
Lower limit	$14 - 1.5 \times 5 = 6.5$
Upper limit	$19 + 1.5 \times 5 = 26.5$
Estimated range	$26.5 - 6.5 = 20$
Outlier	32

Box-Whisker plot



2. The following data represent the daily ticket sales at a park during two weeks.

198, 172, 69, 183, 84, 105, 118, 156, 191, 92, 149, 118, 147, 133

- Construct a Stem-Leaf diagram to illustrate the data.
- Use your diagram to find the median and mode of the data.
- Estimate the nature of the given data.
- Find the semi-range of the above-mentioned data.
- Find the percentage of the daily tickets selling between 90 to 140.

Stem	Leaf
6	9
8	4
9	2
10	5
11	8, 8
13	3
14	7, 9
15	6
17	2
18	3
19	1, 8

Key: 19|1 → 191

Median	$\frac{133 + 147}{2} = 140$
Mode	118
Q1	105
Q3	172
Q1~Median	35
Median~Q3	32
Nature	Left-skewed
Semi-range	$\frac{198 - 69}{2} = 64.5$

Percentage of the daily tickets selling between 90 to 140 is $\frac{5}{14} \times 100\% = 35.71\%$.

3. The following data represent the daily ticket sales at a park by two sellers.

Seller A: 194, 105, 168, 118, 181, 133, 157, 172, 189, 199, 146

Seller B: 102, 117, 133, 145, 110, 128, 99, 156, 120, 140, 111, 153

- Construct a back-to-back Stem-Leaf plot to illustrate the data.
- Plot the Box-Whisker plot corresponding to the Stem-Leaf plot in (a).
- Discuss the consistency and nature of the provided data.
- Use your Stem-Leaf plot to find the modes of the sellers.

Seller B	Stem	Seller A
9	9	
2	10	5
7, 1, 0	11	8
8, 0	12	
3	13	3
5, 0	14	6
6, 3	15	7
	16	8
	17	2
	18	1, 9
	19	4, 9

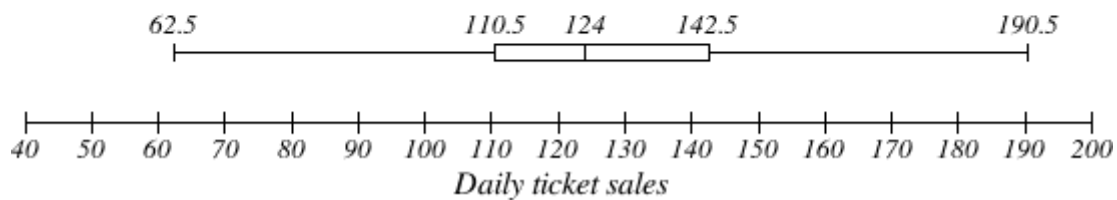
Key: 0|14|6 $\rightarrow A = 146$ & $B = 140$

	Seller A	Seller B
Median	168	$\frac{120 + 128}{2} = 124$
Q1	133	$\frac{110 + 111}{2} = 110.5$
Q3	189	$\frac{140 + 145}{2} = 142.5$
IQR	56	32
LL	$133 - 1.5 \times 56 = 49$	$110.5 - 1.5 \times 32 = 62.5$
UL	$189 + 1.5 \times 56 = 273$	$142.5 + 15 \times 32 = 190.5$

Seller A



Seller B



According to the above Box-Whisker plot, Seller B is more consistent. Daily ticket sales by Seller A are left-skewed, whereas it is right-skewed for Seller B.

There is no mode exists for either of the sellers.

4. The following data represent the daily ticket sales at a park by two sellers.

Seller A: 194, 105, 168, 118, 181, 133, 157, 172, 189, 199, 146

Seller B: 102, 117, 133, 145, 110, 128, 99, 156, 120, 140, 111, 153

(a). Construct a back-to-back Stem-Leaf plot to illustrate the data.

(b). Plot the Box-Whisker plot corresponding to the Stem-Leaf plot in (a).

Seller B	Stem	Seller A
9	9	
2	10	5
7, 1, 0	11	8
8, 0	12	
3	13	3
5, 0	14	6
6, 3	15	7
	16	8
	17	2
	18	1, 9
	19	4, 9

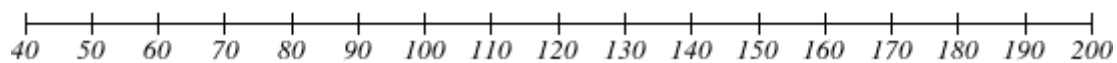
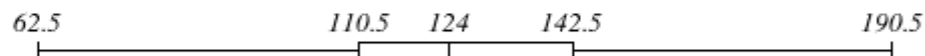
Key: 0|14|6 $\rightarrow$ A = 146 & B = 140

	Seller A	Seller B
Median	168	$\frac{120 + 128}{2} = 124$
Q1	133	$\frac{110 + 111}{2} = 110.5$
Q3	189	$\frac{140 + 145}{2} = 142.5$
IQR	56	32
LL	$133 - 1.5 \times 56 = 49$	$110.5 - 1.5 \times 32 = 62.5$
UL	$189 + 1.5 \times 56 = 273$	$142.5 + 15 \times 32 = 190.5$

Seller A



Seller B



Daily ticket sales

5. A sports fan compiled the Seattle Supersonics' scores from the first 20 games of their first season (1967-68) and their last season (2007-08). The back-to-back stem-and-leaf plot below shows the scores.

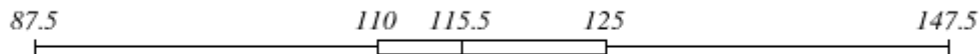
1967-68 Season		2007-08 Season
	7	6
	8	4
6, 4	9	3, 4, 5, 5, 6, 8, 8, 9, 9
9, 5, 0	10	1, 1, 1, 3, 3, 4, 4, 8
8, 8, 7, 6, 5, 4, 4, 2, 1	11	
9, 5, 5	12	6
7, 2, 0	13	

Key: 5|12|6 → 126 (2007 – 08) & 125 (1967 – 68)

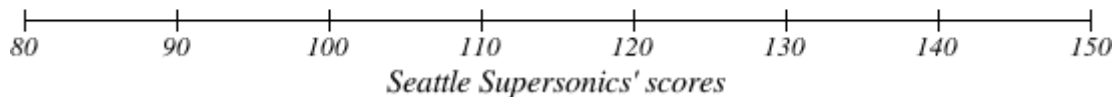
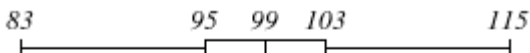
- (a). Construct a Box-Whisker plot according to the given Stem-Leaf plot and identify that which session is more consistent.
- (b). Which is the mode of the scores in the session 2007-08?
- (c). Find the percentage of scores in the session 1967-68 is at least 119?
- (d). If the target is to maximize the scores, which session is better?

	1967-68	2007-08
Mode		101
Median	$\frac{115 + 116}{2} = 115.5$	$\frac{99 + 99}{2} = 99$
Q1	$\frac{109 + 111}{2} = 110$	$\frac{95 + 95}{2} = 95$
Q3	$\frac{125 + 125}{2} = 125$	$\frac{103 + 103}{2} = 103$
IQR	15	8
LL	$110 - 1.5 \times 15 = 87.5$	$95 - 1.5 \times 8 = 83$
UL	$125 + 1.5 \times 15 = 147.5$	$103 + 1.5 \times 8 = 115$

1967-68



2007-08

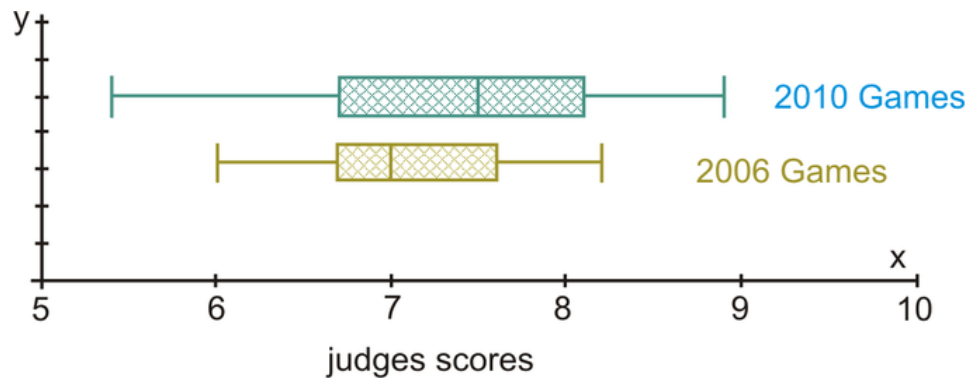


Percentage of scores in the session 1967-68 is at least 119 is $\frac{6}{20} \times 100\% = 30\%$.

According to the above table, the session 1967-68 is better for maximizing.

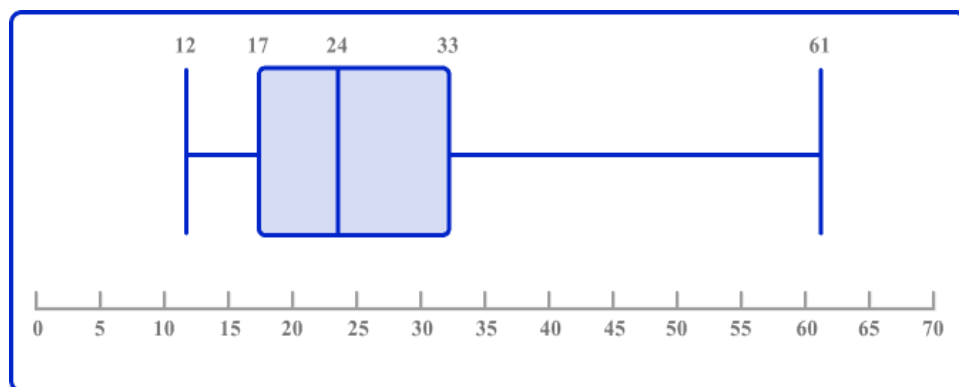
From the above Box-Whisker plot, it is evident that the session 2007-08 is more consistent.

6. According to the following Box-Whisker plot, comment on the central tendency and consistency of the maximizing the scores of the athletes.



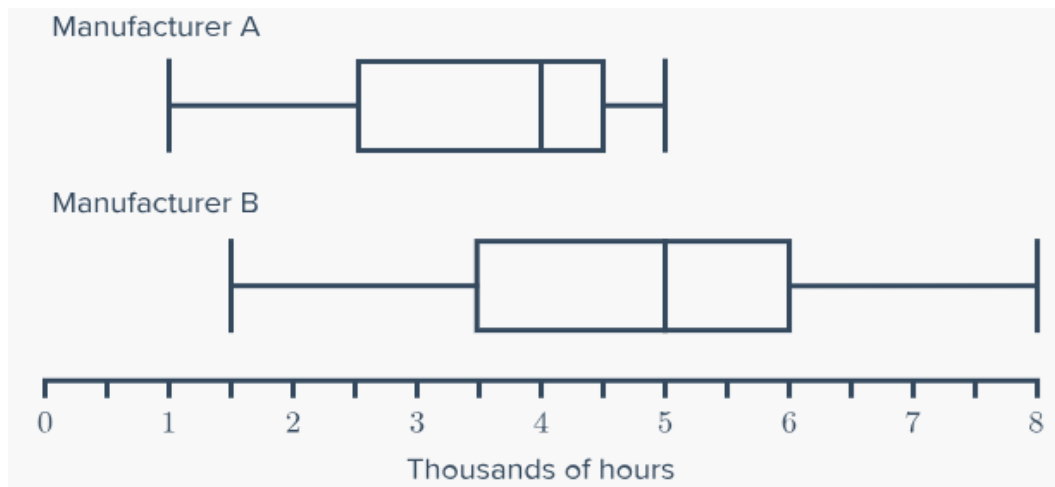
According to the above Box-Whisker plot, the athletes were better in 2010 for maximizing, whereas in 2006 they were more consistent.

7. According to the following Box-Whisker plot write the relation among the Mode, Median, and Mean. Also, find the semi-range.



According to the above Box-Whisker plot, the nature of the data is right-skewed. So, the required relation is $Mo < Me < AM$. Also, the semi-range is $\frac{61-12}{2} = 24.5$.

8. According to the following Box-Whisker plot, comment on the central tendency and consistency of the maximizing the yearly working hours of the employees. Also, find the semi-range for manufacturer A and the nature for manufacturer B. [4]



According to the above Box-Whisker plot, the Manufacturer B is better for maximizing, whereas in Manufacturer A is more consistent.

The semi-range of Manufacturer A is $\frac{5000-1000}{2} = 2000$, whereas nature of the Manufacturer B is left-skewed.